Surrogate-Assisted Modeling and Robust Optimization of a Micro Gas Turbine Plant With Carbon Capture

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Goal of the Modelling Process

The main goal of the modeling process is to perform a Robust Design Optimization (RDO) of a micro gas turbine (mGT) with Exhaust Gas Recirculation (EGR) with carbon capture (CC) plant

Model Identification

	Classification	Reason
Static/Dynamic	STATIC	Steady-state simulations
Lumped/Distributed	DISTRIBUTED	Columns are distributed (RadFrac)
Deterministic/Stochastic	STOCHASTIC	Gaussian inputs, Monte Carlo, RDO
Inputs	7 variables	Power, Temperatures, O_2 Concentration…
Outputs	μ, σ	Mean efficiency, standard deviation
Main Goal	RDO	Robust optimization: maximize μ, minimize σ

The model is Black

Because Gaussian Process Regression (GPR) surrogate models, which are used in this model, are purely statistical - they have polynomial and Gaussian process functions, but no physics equations.

Input -> Output

Applications

- Optimization of operations
- Robust of optimization
- Emissions modeling

Mechanisms and Equations

- Heat transfer
- Chemical reactions
- Thermodynamic processes
- Fluid mechanics (flow)
- Phase changes
- Mechanical process

$$\frac{\dot{m}\sqrt{\mathrm{TIT}}}{\mathrm{PIT}} = A\sqrt{\frac{k}{R}\left(\frac{2}{k+1}\right)^{\frac{k+1}{k-1}}} = \text{constant}$$

$$\frac{\eta}{\eta_*} = \frac{k-1}{k_*-1}\sqrt{\frac{k+1}{k_*+1}\frac{1-1/\pi^{\gamma_*}}{1-1/\pi^{\gamma}}} \quad \text{where} \quad \gamma = \frac{k-1}{k}$$

$$y(\mathbf{x}) = \sum_{i=0}^{p} \beta_i f_i(\mathbf{x}) + z(\mathbf{x}) = \boldsymbol{\beta}^{\mathrm{T}}\mathbf{f}(\mathbf{x}) + z(\mathbf{x})$$

Quality of results and validation

- The researchers compare the stochastic with deterministic results to validate the model, which is reliable and accurate
- The quality of the results is checked by comparing surrogate model results with ASPEN PLUS simulations, showing very small variation

Seven-step modelling Procedure

- The researchers follow every step of the seven-step modelling Procedure